Safeway Mobile App: Usability Report

Project 1

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Introduction: Product Selection (Giovanni)

For our midterm UI evaluation project, we chose to evaluate the Safeway mobile app. Safeway is a known grocery store that has many locations across the country. While the team considered other grocery mobile apps, we decided to focus on Safeway App, as we identified extensive issues that would create friction in a user's daily shopping experience. The goal of this report is to explore the app's effectiveness, efficiency, and ease of use, to ensure we can address user's needs and frustrations to enhance their user experience.

Product Description (Giovanni & Zaijing)

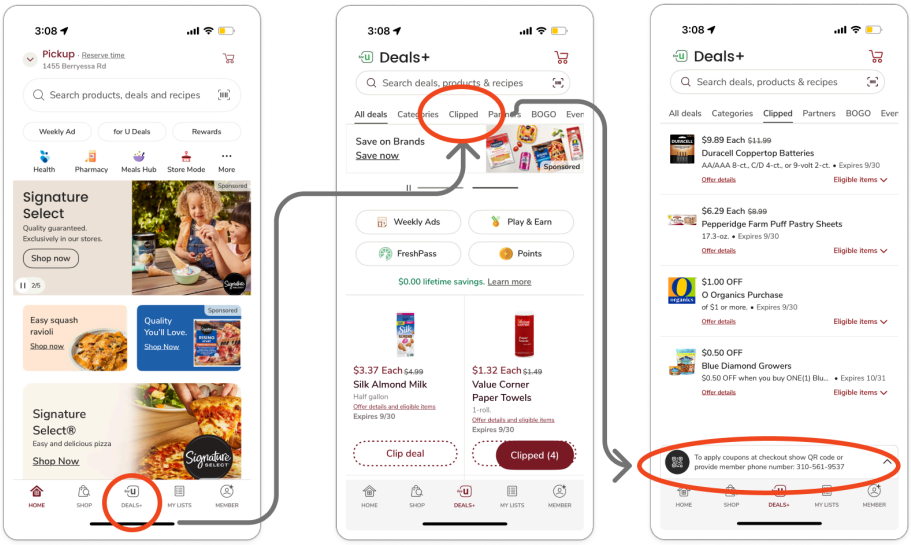
Safeway is a subsidiary of Albertsons Companies. Its mission is to earn customer loyalty by anticipating and fulfilling shoppers' needs through high-quality products, enhanced shopping experiences, and customer-focused service. To meet the demands of the digital era, they launched the Safeway app, which intends to support this mission by offering digital couponing, meal planning, and access to reliable ingredients. Through the app, users are able to order groceries for pickup and delivery, access coupons for different products, scroll through recommended seasonal food recommendations, and purchase ingredients based on recipes. The Safeway application is available on both IOS and Android, which can be downloaded at no cost.

Interface Evaluation

Interface Evaluation Methodology (Giovanni & Zaijing)

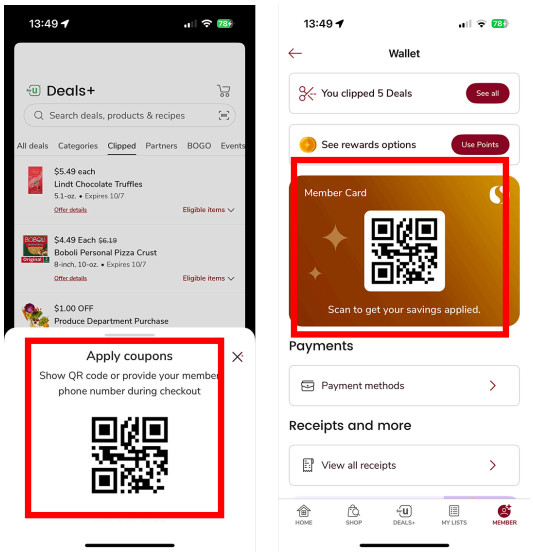
We conducted an informal heuristic evaluation to assess the user interface of the Safeway App. An application walkthrough was conducted by 3 members of the team to identify inconsistencies and opportunities for UI improvement. We assessed and critiqued UI elements and the interface in accordance to the 8 Golden Rules of Interface Design, Fitt's Law, Gestalt, and Hick-Hyman's Law. We ranked them according to the level of importance.

Figure 1: Rewards QR Code for in-person couponing



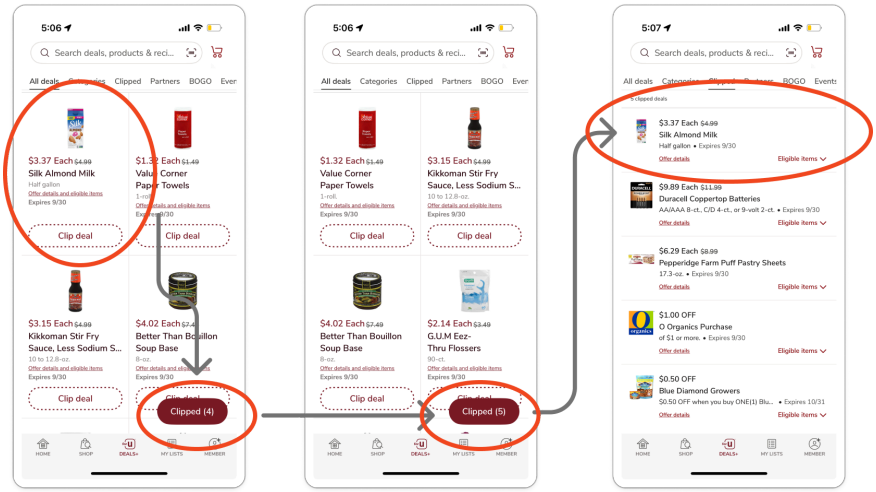
Assessed Issue & Guidelines / Principles Review	Recommendations
Priority 1: The rewards QR code to apply coupons for in-person purchases **(Figure 1)** is difficult to find. Users must dig into the app at checkout to ensure they receive their discounts and points. Couponing and rewards systems are a primary function of many grocery apps. ***Violation of Fitts'Law*:** The UI element is small and placed at the bottom of the screen. Scanning rewards should be larger and quicker to access. ***Strive for Consistency:*** The app does not demonstrate consistency in making one of its primary features prominent, unlike its competitors. ***Figure & Ground:*** The barcode is visually minimized at the bottom, so it does not appear to be the main object of attention.	Make the rewards QR barcode more visually prominent so users can easily recognize and identify the feature. Elevate its placement in the menu hierarchy to ensure quick access at checkout, reducing frustration, time on task, and unnecessary clicks.

Figure 2: QR code



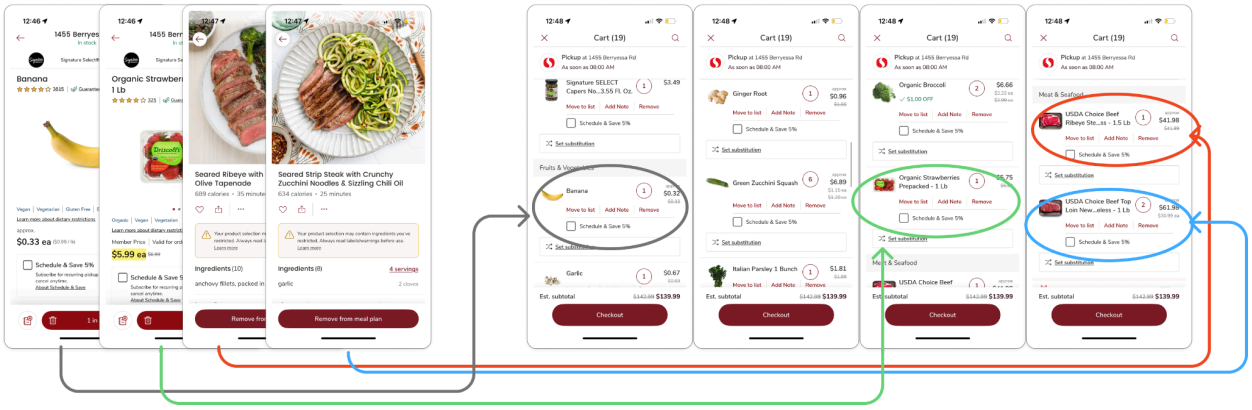
Assessed Issue & Guidelines / Principles Review	Recommendations
Priority 1: This app displays two QR codes in multiple locations (Deal page and Wallet) but serves the same function which is to apply coupons. But the visual and the contexts are differ which may make users confused and think they are separate different features. (**Figure 2**) ***Consistency:*** This design violates the principle of consistency by displaying the member QR code in different pages with different visual and contexts. Although both serve the same function, the inconsistent display makes users think they are different features, and this leading to confusion. ***Reduce Short-Term Memory :*** Users should remember at least one path to access this QR code. Instead of visually surfacing a direct point like only click members will show the code, this design increases memory demand. During checkout, users need a fast and stress-free way to access the QR code.	Move the QR code from the Wallet to the Member page, so users can access it with fewer clicks and without switching between multiple pages. In addition, unify the visual design and context of the QR code across the app to ensure that users clearly understand it is the same function.

Figure 3: Clipped Coupon Feedback



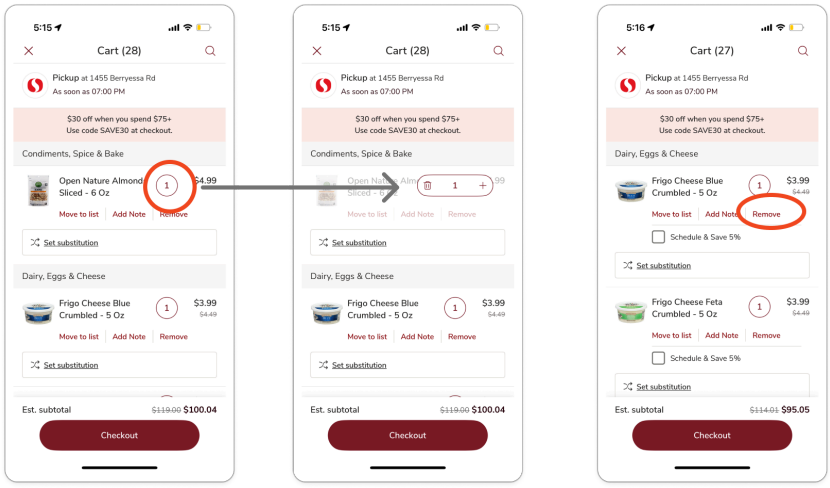
Assessed Issue & Guidelines / Principles Review	Recommendations
Priority 1: When users clip coupons, the process creates uncertainty and confusion, disrupting their shopping experience **(Figure 3)**. The system does not provide clear feedback to confirm that a coupon has been successfully added. ***Offer Informative Feedback (Golden Rules):*** The interface fails to notify users when their action is completed, leaving them unsure if the coupon was clipped. ***Recognition Rather than Recall:*** Users must remember how many coupons they had clipped before in order to notice the counter update. Without explicit confirmation, they cannot easily recognize that the action was successful. ***Design Dialogs to Yield Closure (8 Golden Rules):*** Clipping a coupon is only the first step in obtaining discounts, yet the system provides no guidance or sense of completion to help users understand what comes next.	Provide clear, immediate feedback when users click on "Clip Deal" by displaying a confirmation state. Surface a contextual next step to guide users through the couponing process.

Figure 4: Shopping Cart Grouping



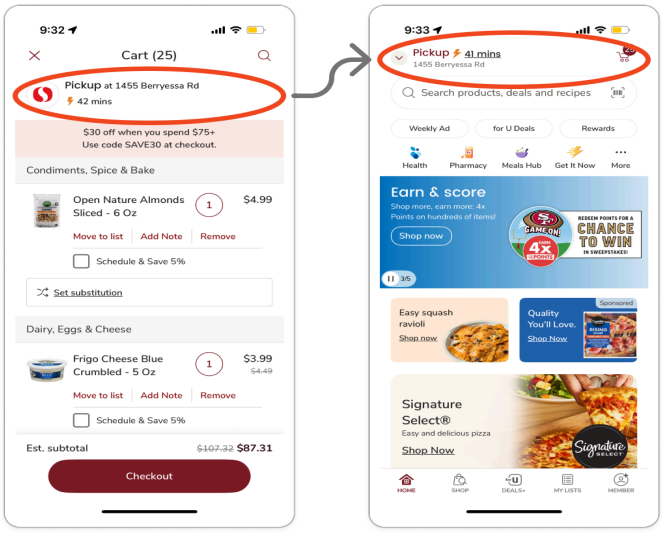
Assessed Issue & Guidelines / Principles Review	Recommendations
Priority 1: Identifying items in the shopping cart increases cognitive load and frustration for users when they add both à la carte items and meal plans **(Figure 4)**. ***Reduce Short-Term Memory Load (Golden Rules):*** The design forces users to hold in memory which ingredients belong to which meal, instead of showing it visually. This is especially problematic when some plans include 8+ ingredients, exceeding the limits of short-term memory. ***Gestalt Principle of Proximity:*** The cart groups items by type (grains, fruits & veggies, meats), which misrepresents the mapping users expect. Based on their previous actions, users anticipate seeing items grouped by a la carte selections or by meal plan. ***Error Prevention & Flexibility and Efficiency of Use (Nielsen's Heuristics):*** Users may accidentally remove the wrong item from a meal plan or a la carte selection due to the confusing organization. There is also no efficient way to manage meal plans within the cart; users must backtrack to identify items manually or restart the entire process.	Introduce a tabbed view that allows users to toggle between a la carte items and meal plans. This layered organization preserves the existing grouping by type (grains, fruits & veggies, meats) while adding a higher-level distinction that aligns with user expectations. By surfacing meal plans as clear groupings, users can more easily manage, review, or remove items without unnecessary cognitive load.

Figure 5: Removing Items UI Elements



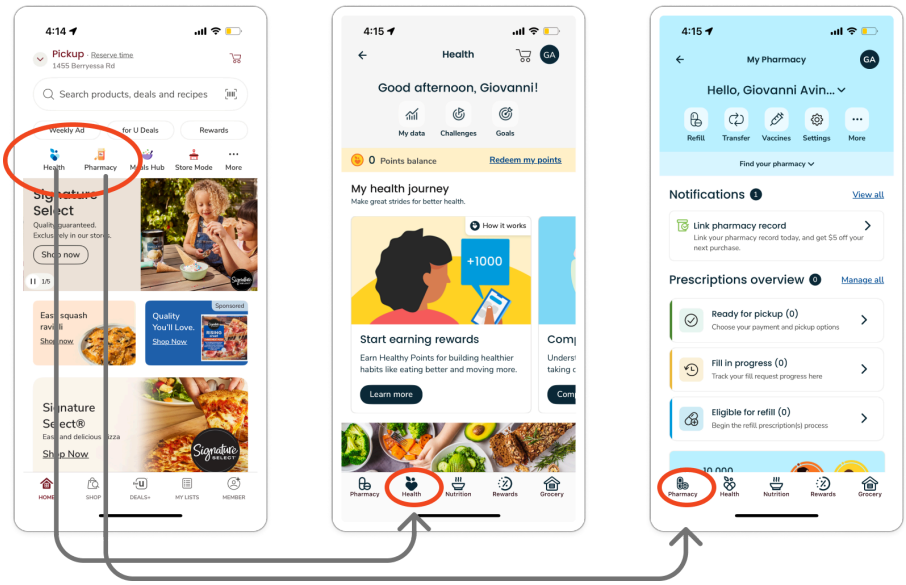
Assessed Issue & Guidelines / Principles Review	Recommendations
Priority 2: The quantity adjustment control **(Figure 5)** is represented only by a numeral, which *lacks clear affordance*. ***8 Golden Rules of Interface Design:*** Strive for Consistency is violated by failing to follow standard UI patterns for quantity adjustment. Prevent Errors are also violated, as the remove button is placed very close to the quantity control, increasing the risk of accidental item deletion. ***Gestalt Principle of Proximity:*** The placement of the remove button directly beneath the quantity control visually groups them, implying a functional relationship that does not exist and potentially misleading users.	Replace the standalone numeral with clear quantity adjustment controls, such as +/– buttons, to align with established interaction patterns and affordances. Integrate an X icon for item removal or a checkbox to support multi-item deletion. This reduces ambiguity, prevents errors, and creates a more intuitive and efficient cart management experience.

Figure 6: Inflexible Address Change



Assessed Issue & Guidelines / Principles Review	Recommendations
Priority 2: Users cannot change the store address directly from the shopping cart. Clicking the address redirects them to the home screen forcing them to adjust the address there and then navigate back to the checkout **(Figure 6)**. *This workflow inefficiency violates several Nielsen's Heuristics and one of the golden rules.* ***User Control & Freedom:*** Users are unable to change the store address directly in the cart view, so they are forced to leave the checkout flow adding unnecessary steps ***Flexibility & Efficiency of Use:*** Changing store location is a common action in a shopping experience, instead of having the functionality accessible at any moment, users have to backtrack to proceed with the action. ***Cater to Universal Usability:*** The system failed to design for a wide range of users. Shoppers who are new or under time constraints will become frustrated and possibly abandon their shopping goal at checkout.	Enable users to update their store location directly within key points of the checkout flow. Providing in-context address controls ensures a seamless shopping experience, reduces unnecessary navigation, and supports task completion without disruptions.

Figure 7: Redundant Icons Leading to Same Destination



Assessed Issue & Guidelines / Principles Review	Recommendations
Priority 3: The Health and Pharmacy icons **(Figure 7)** are redundant, as they lead users to the same destination, differing only by a tab. ***Violation of Consistency & Standards:*** Users expect different functionalities from different icons. Having Health and Pharmacy direct to the same page creates confusion and breaks consistency. ***Aesthetic and Minimalist Design:*** Redundant entry points increase complexity and add unnecessary clutter to the interface. ***Similarity:*** Users naturally group elements they perceive as alike. Here, the icons break expectations by presenting visual differences that do not correspond to functional differences.	Consolidate the functionality into a single icon and relabel it as Wellness, creating a unified entry point that clearly represents both pharmacy and health services.

Features Evaluated for Usability Testing (Giovanni)

With the results of our initial evaluation, the team decided to conduct usability testing sessions with users on three main features within the Safeway App.

1. **Clipping & Applying Online Coupons:** Find a coupon, clipping the coupon, and successfully finding an eligible item to add to the shopping cart.
2. **Reviewing Shopping Cart:** Adding a la carte and meal plan items , and successfully removing 1 meal plan
3. **Member Card:** Finding and scanning member card for in-store purchases

Usability Goals (Giovanni)

From our evaluation, we identified four critical usability goals to focus our research on the Safeway App:

1. **Effectiveness:** Determine whether users can successfully achieve their intended goals within the app.
2. **Efficiency:** Evaluate if users can perform tasks quickly and with minimal friction.
3. **Ease of Use:** Assess whether key features are intuitive and easily discoverable.
4. **Satisfaction:** Identify pain points and opportunities to improve the overall user experience.

To further guide the team on evaluating the application, we formulated research questions based on our goals to assess how well the Safeway App performs:

- **RQ1:** How effectively can customers locate, clip, and apply coupons, and find eligible items to add to their cart?
- **RQ2:** What challenges do customers face when managing their shopping carts when adding a la carte items and meal plans?
- **RQ3:** How easily can customers access their member card to scan during in-store purchases, and how does this process influence their overall satisfaction with the app?

Tasks & Their Process Flows (Giovanni)

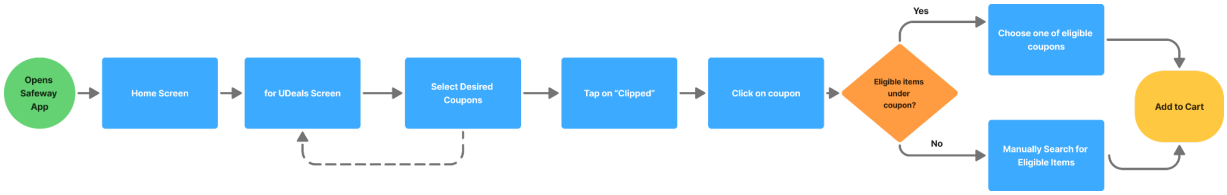
During our walkthrough of the selected features, the team conducted a task analysis to map the flows required to achieve each goal. This analysis allowed us to redefine tasks in ways that reduce knowledge requirements and cognitive effort. We then outlined the expected user flows for each task to evaluate the likely sequence of actions during usability testing.

Feature #1: Clipping & Applying Online Coupons

Tasks:

1. Open Safeway App
2. Navigate to for for U Deals screen (click either on icon or navigation button)

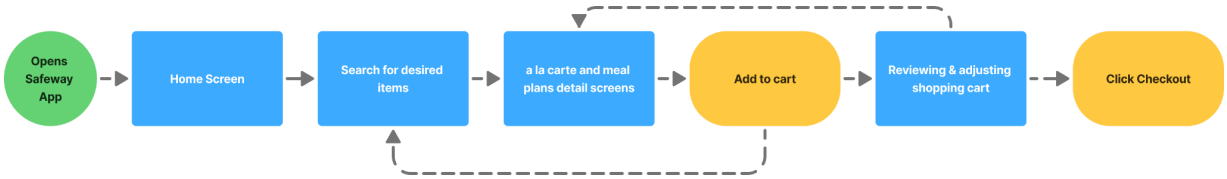
3. Selected desired coupon
4. Navigate to “Clipped” section in the For U Deals section
5. Select desired coupons
6. Navigate to “clipped” section
7. Click on the coupon
8. Click on “eligible items”
9. Choose one of the eligible items
10. Add to cart



Feature #2: Shopping Cart

Tasks:

1. Open Safeway App
2. Click on shop or search for items freely
3. Add user’s preferred items to cart (a la carte & meal plans)
4. Review and adjust shopping cart
5. Click Checkout



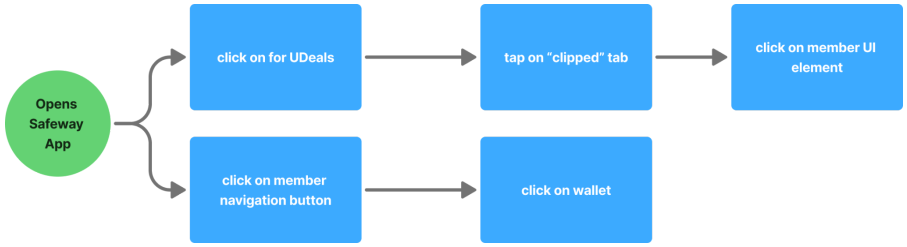
Feature #3: Member Card

Method 1 Tasks:

1. Open Safeway App
2. Click on for UDeals
3. Click on “Clipped” tab
4. Click on bar code UI element at bottom of screen

Method 2 Tasks:

1. Open Safeway App
2. Click on member navigation button
3. Click on Wallet



Competitive Analysis (Jonathan)

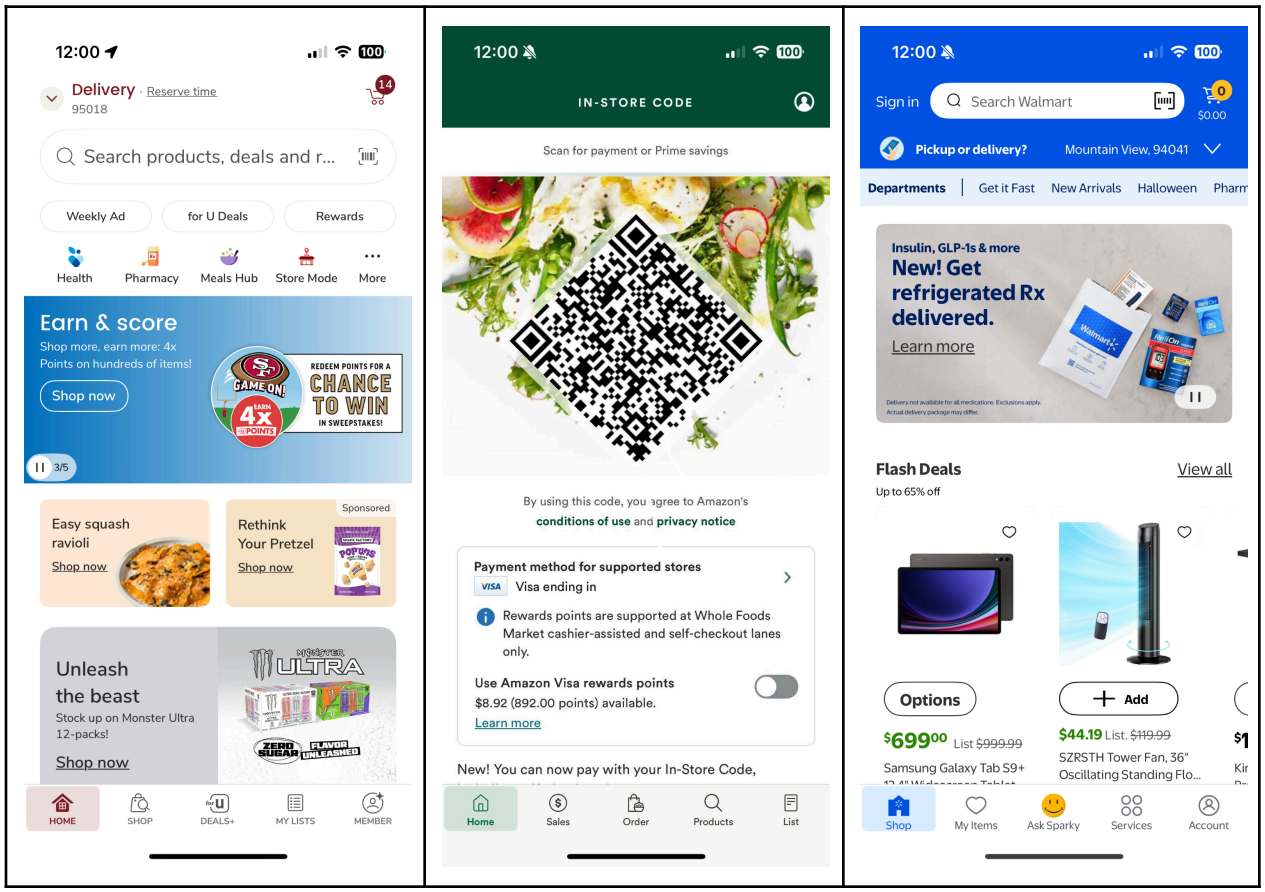
We compared the Safeway application against two similar grocery applications, Whole Foods and Walmart, in order to conduct a competitive analysis and analyze the market competition of Safeway. The table below summarizes our findings:

Contents	**Safeway**	**Whole Foods**	**Walmart**
Purpose	A mobile application for grocery shopping at Safeway stores, supporting pickup and delivery services.	A mobile application that integrates with Amazon Prime to provide access to discounts and store information at Whole Foods.	A mobile application for shopping at Walmart that combines grocery and retail store selection
User Interface	• Launch page: "Home" tab that links to different functions in the application as well as recommended meals and products • Navigation bar on the bottom with HOME, SHOP, DEALS+, MY LISTS, and	• Launch page: "Home" tab with QR code meant to be scanned in store for linking to Amazon account • Navigation Bar on the bottom with Home, Sales, Order, Products, and List tabs • User's personal information is	• Launch page: "Shop" tab that features a rotation of new products and deals • Navigation Bar on the bottom with Shop, My Items, Ask Sparky, Services, and Account tabs

	MEMBER tabs • User's personal information is located in the MEMBER tab in the bottom bar • Search Bar includes barcode scanner for in-person shopping • Digital Coupon "Clipping" • Integration with pharmacy and prescription service • "+Add" buttons to quickly add items to cart • Pickup and Delivery options	found by clicking on the Profile icon when browsing the Home tab • Pickup and Delivery options • Order tab directs users to "Grocery Pickup & Delivery," "Catering," and "Shipped to You"	• User's personal information is located in the Account tab in the bottom bar • Search Bar includes barcode scanner for in-person shopping • Pickup and Delivery options • AI Assistant Sparky • Horizontal carousels that display 4 products at a time
UI Elements	• Search bar with barcode scanner and suggestions • Horizontal carousel of featured / promotional products • Shopping cart icon with persistent cart access • Product reviews: star ratings and user review count • Coupons / "clip deal" badges on product cards	• Search bar with suggestions • Shopping cart icon • Deals / "Prime offers" banners, promotions section • Filters & categories: e.g. organic, dietary filters, local brands • Shopping list tab • Store selector • Offline in-store scan code / QR code for Prime	• Search bar with barcode scanner and suggestions • Product reviews: star ratings, user review count, and comments • Shopping cart icon with persistent cart access • Horizontal carousels of products • Filters &

	• "Store Mode" aisle locator • My Lists tab for separate shopping lists • Filters & sorting: e.g. by price, category, availability • Rewards with points display	discounts • Profile / account settings for preferences, payment, address	sorting (price, rating, department, availability) • Banner / hero promotions on home screen • Vendor / "fulfilled by Walmart" tags on product cards
Advantages	• In the dedicated Deals tab users can browse weekly ads and manage clip coupons from within the same interface. • "Meals Hub" allows users to browse through complete recipes and add multiple items to their shopping cart at once • Shopping list feature helps users keep track of specific products. Users are also able to take a picture of physical lists to import items	• Product selection allows users to shop items by local availability • Integration with Amazon Prime gives a seamless link between the digital and physical shopping experience.	• Dedicated Services tab separates grocery shopping features from separate ones • Color Palette clearly distinguishes UI Elements • Search & filtering system helps users narrow down large catalogue of products
Disadvantages	• Coupon Clipping is unclear • Cluttered product information- multiple sets of words are cut off	• Unclear if items can be added to shopping cart • Search bar offers minimal suggestions when	• Unclear when links send user to different pages in application or to the Walmart website, ex:

	when browsing products • Adding items to Shopping List is easy to mess up • The add to shopping list icon is too small (Fitts Law)	looking for products • Application is slow to load between pages • It is not always possible for users to go back to previous screens	clicking on “Custom Cakes” takes users out of application • Visual clutter in product listings can be overwhelming for user selection



Competitive Findings

Our comparative analysis of the Safeway, Whole Foods, and Walmart mobile applications reveals key similarities as well as distinct differences in user experience. The Whole Foods app offers a minimal interface that integrates seamlessly with Amazon Prime, but this dependency limits functionality for users who prefer to shop independently of Amazon. Its design emphasizes in-store shopping by featuring a personal QR code on the Home tab, whereas Safeway’s

equivalent feature, the “Member Card,” is buried two clicks deep in the “Wallet” within the “MEMBER” tab. In contrast, Walmart, delivers a more consistent and intuitive interface, with clear navigation and well-structured product organization. However, newer features such as the AI assistant “Ask Sparky” deliver inconsistent results and occupy valuable space in the bottom bar of the screen. Based on these comparisons, Safeway would benefit from refining its interface layout and product navigation to create a more streamlined and transparent user journey, ultimately enhancing customer satisfaction, engagement, and sales.

User Profiling (Zaijing)

Our target users are primarily college students who have a limited budget and are interested in saving money and time when shopping for groceries. They frequently use the app to clip coupons, plan meals and shop online for delivery or visit stores for pick up. According to heuristic evaluation, we identified some potential problems, such as coupons lacking feedback and coupons violating consistency… Using these findings, we created persona, which helped us understand the target users and some assumed pain points, and also supported the setup of scenarios and tasks for the usability test. The next step is to use the usability test to verify whether these pain points actually exist.



Emily

Age: 21

Gender: Female

Occupation: Design Student

Location: San Jose

"I just want to save money and cook something quick without spending too much time searching in the app"

Motivation:

- Save money with coupons
- Save time by shopping online
- Discover and try easy recipes

Goal:

- Quickly clip and use coupons online to lower the grocery costs
- Use one-click purchase all ingredients in Meal Hub to cook simple meals

Frustrations:

- When she accidentally chooses the wrong meal in Meal Hub, the system automatically adds all ingredients, making it difficult to distinguish which items she actually needs.
- After clipping coupons, there is no clear confirmation, so she is unsure if the coupon was successful.

Emily is a graduate design student living in an off-campus apartment with roommates. She often looks for coupons to save money because her living expenses are limited. She prefers to use the delivery service instead of picking up in stores, because she needs to spend most of her day attending classes and doing projects. In her free time, she likes to explore simple recipes on Meal Hubs, and she often uses the one-click purchase function to order all the ingredients online.



Kevin

Age: 32

Gender: Male

Occupation: single father with two kids

Location: San Jose

"When I do my monthly grocery run, I want to be able to adjust my shopping cart as quick as possible"

Motivation:

- Save money with coupons
- Ensure food and drinks meet nutrition and quality standards
- Explore and discover different menu options for variety

Goal:

- Clip coupons easily to find and use them during checkout
- Explore and try some recipes from the app
- Search up items quickly

Frustrations:

- Sometimes cannot find or access the QR code for coupons when paying in-store.
- Having to navigate to another page just to apply a coupon for an eligible item
- When he can't find items using the search bar, and when the algorithm doesn't adjust while typing
- Coupon confirmation feedback is limited, making it hard to know which ones are active at checkout

Kevin is a 32-year-old single father living in San Jose with his two children. Managing both work and parenting keeps him very busy, so he values tools that can save him time and money. He often uses the Safeway app to clip coupons before going to the store, which helps him reduce grocery costs. Kevin also pays close attention to nutrition and variety when planning meals for his kids. He occasionally explores new recipes in the app's Meal Hub to prepare healthy, creative meals that his children will enjoy.

Usability Metrics (Giovanni)

We conducted usability tests to further evaluate the Safeway App. Participants were asked to think aloud while they performed the tasks. The following usability performance measures were utilized to focus on our key goals: effectiveness, efficiency, and ease of use.

- **Task Completion Time**: Time the participant took to complete the task
- **Completion Success:** If the participant successfully completed the task
- **Error Rate:** Each tap/click the user did that was not directed towards the goal
- **Number of Clicks:** Amount of clicks the participant did to complete the task

The team also collected qualitative data during the usability tests, to have a comprehensive understanding of their experience with the Safeway App. Qualitative measures included:

- **Subjective:** Think aloud feedback and verbal responses from the team's perception probes questions during and after the task
- **Behavioral:** Facial expressions, body language, and other nonverbal cues

Tools (Zaijing)

The following tools to were used to complete the evaluation report:

- **Google Docs**: Used to notetake during usability tests & organize information for the report
- **Figjam/Figma:** To create user flows & annotate Safeway App screens
- **Qualtrics:** Created surveys and generate data visualizations
- **Iphone with Safeway App and Timer:** To conducting usability testing

Techniques (Zaijing)

Survey: To gather quick and cost effective data about user experiences outside of our range
Usability Test:

- **Pre Test questionnaire:** To collect participants' background information which helps establish context and baseline for comparison.
- **User Testing:** To observe participants as they perform core tasks. This reveals usability issues, efficiency problems and points of confusion.
- **Post task questionnaire:** To capture immediate feedback after each task. Participants reflect on the clarity, difficulty and satisfaction of completing specific tasks, and points for improvement.
- **Post Test Questionaire:** To gather overall impressions of the app after completing all tasks. It provides a summary of user satisfaction, perceived usability, and areas for improvement.

Usability Evaluation Based on Usability Metrics

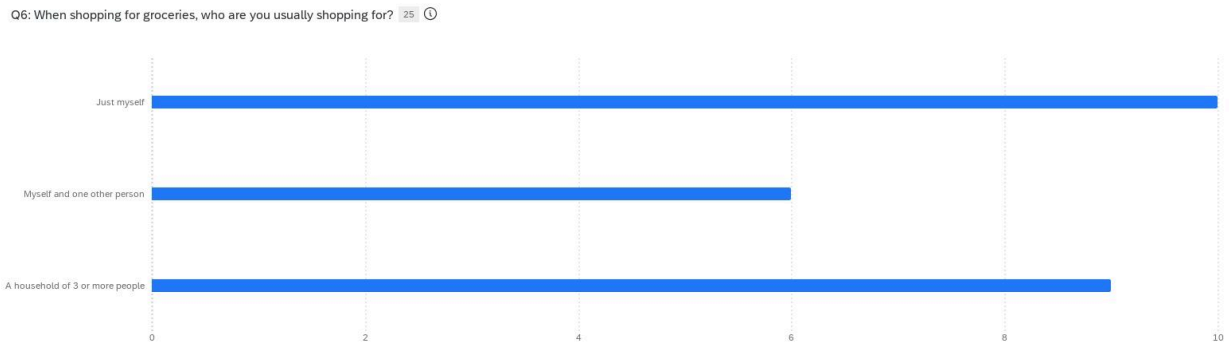
Survey Findings (Giovanni & Jonathan & Zaijing)

In addition to our usability testing, we conducted a survey using the Qualtrics platform to better understand how users interact with the Safeway app. With responses from 25 participants, we examined the user goals, pain points, and overall usability within the application.

Demographics

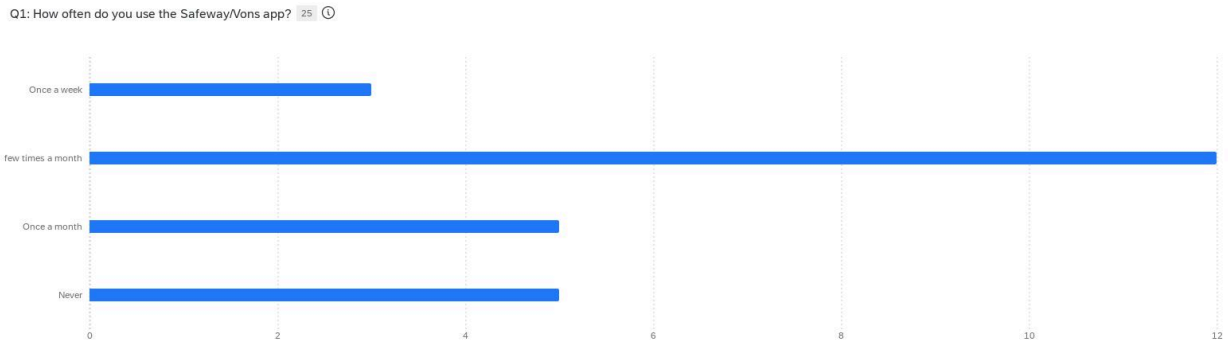
The participants of this survey represented a broad range of grocery shoppers, primarily consisting of individual users and small households:

- 40% shopped alone
- 36% shopped for households of three or more
- 24% with one other person.



In terms of app engagement, participants demonstrated varied levels of usage:

- 48% used the app a few times a month
- 20% had the app installed but rarely used it
- 20% used it once a month
- 12% used it once a week

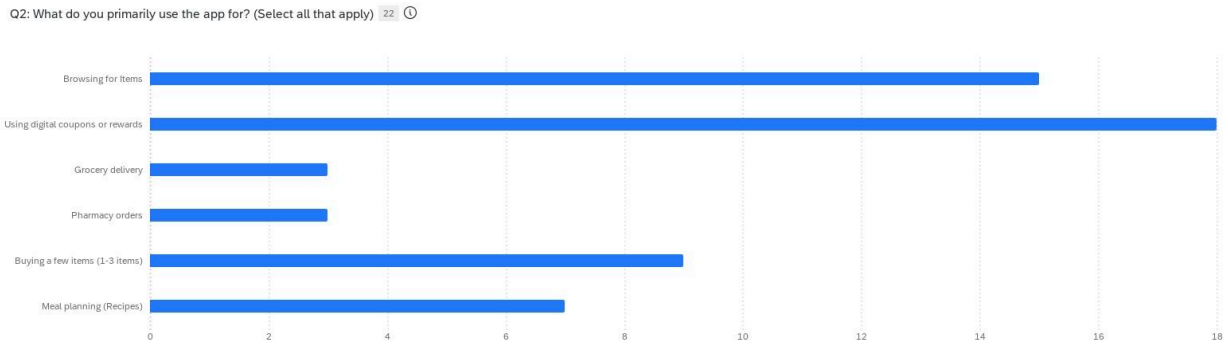


This sample of regular and occasional users provides a balance between regular users of the app and infrequent users, who open it mainly for time-sensitive deals or one-off coupon searches. The responses suggest that, while people may recognize the value of the app, they do not integrate it fully into their grocery routine, likely due to the frustrations we observed such as superfluous effort required to locate deals or navigate between features. These patterns reveal an opportunity for Safeway to improve engagement by streamlining frequent-use functions and reducing friction in task completion.

Primary Uses

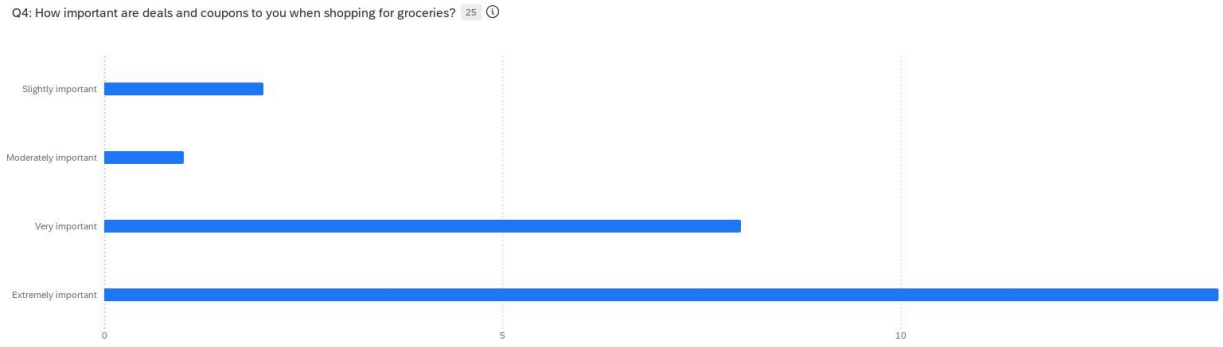
Participants reported using the Safeway/Vons app for a range of shopping-related tasks:

- 82% used it to clip or redeem digital coupons,
- 68% used it to browse for items
- 41% used it to buy few items
- 32% used it for meal planning
- 14% used it for each grocery and pharmacy-related tasks.



Couponing emerged as the most significant motivator for app engagement:

- 56% of participants rated deals and coupons as *very* or *extremely important* to their grocery shopping.



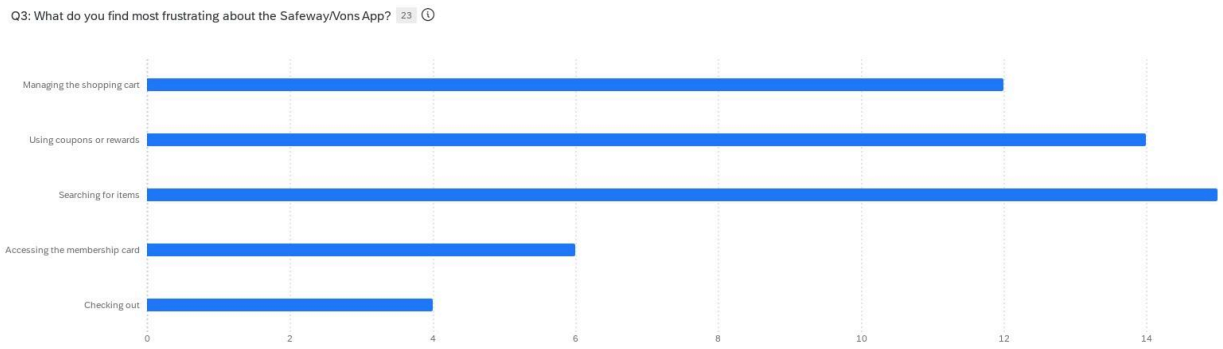
Several respondents also noted that the digital rewards system was a primary reason for downloading and continuing to use the app, even when they did not engage with other functions of the app. This pattern highlights that coupon access and the clipping feature are central to user satisfaction, suggesting that improvements in this area could drive more consistent engagement across multiple demographics.

However, many users voiced that **unclear feedback and multi-step clipping processes** caused frustration. Users often questioned whether coupons had been successfully applied or found the process unnecessarily repetitive.

Common Pain Points

When asked about frustrations, participants most often cited:

- Item search (65%)
- Using coupons or rewards (61%)
- Managing or editing the shopping cart (52%)
- Finding or accessing the member QR card (26%)
- Checkout (17%)

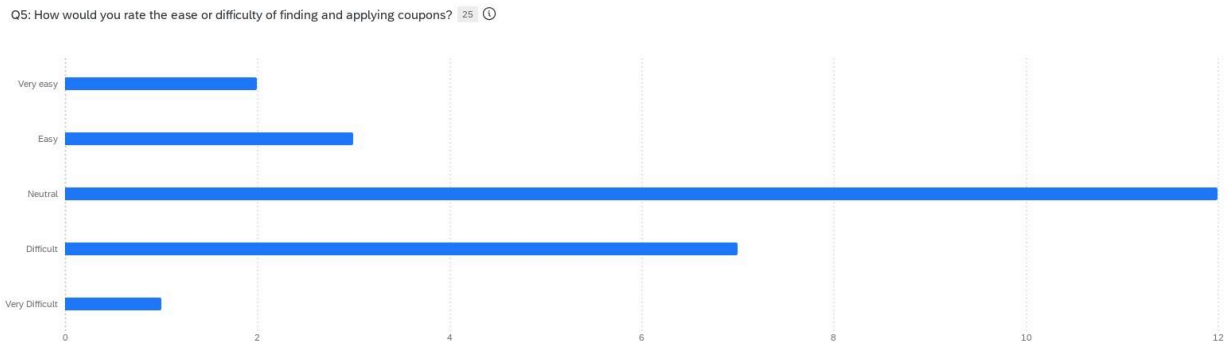


These pain points indicate that users struggle most during task transitions such as searching, moving from browsing products to coupon application or from adding items to checkout. Many participants expressed uncertainty about where to locate key features and reported that repetitive navigation steps disrupted their flow. This indicates that usability issues stem from unnecessary screens or unclear feedback mechanisms that do not follow user expectations.

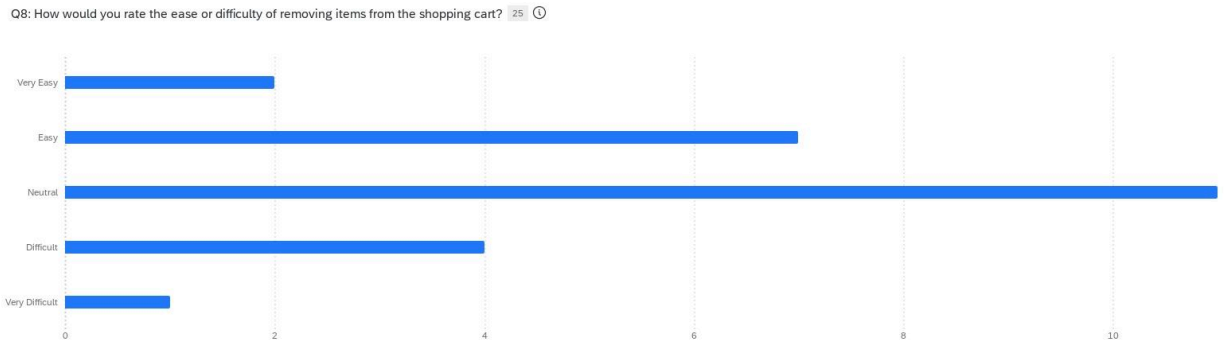
Usability Ratings and Interaction Behaviors

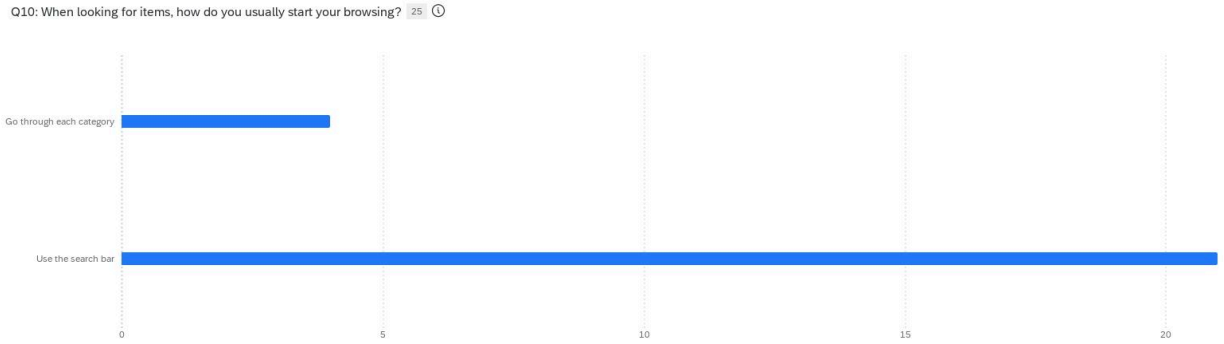
When asked to rate how easy or difficult it was to perform key actions within the app:

- Finding coupons: Nearly half (48%) found the process *neutral* in difficulty, while 28% found it *difficult* or *very difficult*, and only 20% rated it *easy* or *very easy*.
- Access member card: 92% rated as neutral or easy, with 8% reposted difficulty.
- Removing items from the cart: 72% rated it *neutral* or *easy*, though 20% reported difficulty.
- Searching for items: A strong majority (84%) preferred using the *search bar* over browsing categories, suggesting that search optimization plays a crucial role in user efficiency and satisfaction.

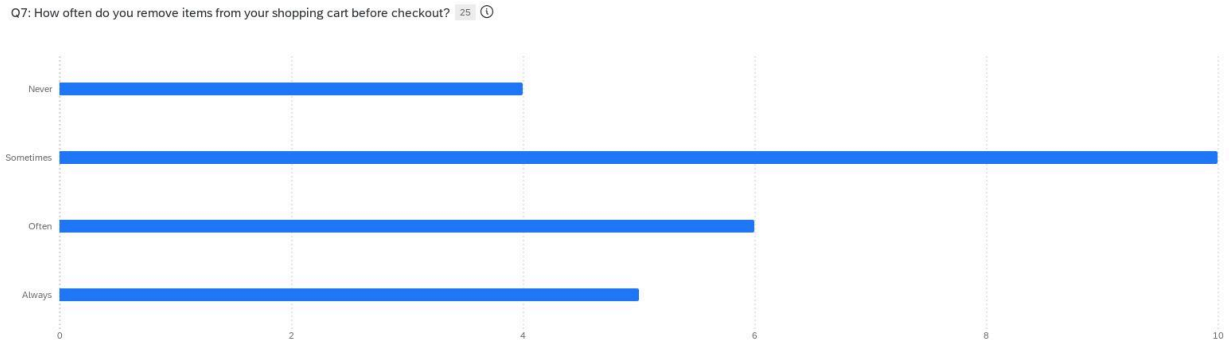








And only 4% of people never remove items from the shopping cart before checkout, which means most people have removed items before they checkout.



Together, these ratings suggest that while basic shopping tasks are manageable, coupon interaction and feature discoverability remain primary barriers to smooth navigation and consistent use.

Overall, the survey demonstrates that most users value the Safeway app for its **digital couponing and browsing features**, but experience significant friction when navigating between tasks or confirming actions. Frequent users rely on the app for deals but encounter barriers in feedback

clarity and feature accessibility. Shopping cart management and member card access also emerged as recurring sources of frustration.

Usability Testing Findings (Giovanni & Zaijing & Jonathan)

The team conducted usability testing to gain a comprehensive understanding of how users interact with the Safeway App. The following section presents the quantitative and observational findings collected from nine participants..

Task 1: Clipping & Applying a Coupon

	P1	P2	P3	P4	P5	P6	P7	P8	P9
Task on Time	30s	26s	40s	45s	50s	1:25	4:58	2:50	4:13
Completion Success?	Y	Y	Y	N	Y	Y	Y	Y	Y
Error Rate (#) **Each tap/click the user did that was not directed towards goal**	0	0	0	3	0	16	21	0	0
Number of Clicks	10	14	18	13	6	23	49	15	26

During Task 1, participants were asked to clip a coupon and apply it to an eligible item. On average, participants completed the task in 1 minute and 51 seconds, requiring an average of 19 clicks with an error rate of 4.4 clicks. Overall, 89% (8 out of 9) participants successfully completed the task. Although most participants were able to complete the task, several expressed frustration and confusion during the process. Many users were irritated by the number of steps required to achieve their goal. As one participant (P2) noted, “I don’t know why I have to go to another page to apply it.”

Task 2: Adjusting Shopping Cart

	P1	P2	P3	P4	P5	P6	P7	P8	P9
Task on Time	5:10	41s	55s	2:10	48s	4:12	3:51	2:50	1:54
Completion Success?	Y	N	N	N	N	Y	Y	Y	Y

Error Rate (#) **Each tap the user did that was not directed towards goal**	50	10	9	8	4	10	4	0	1
Number of Clicks	48	10	11	9	7	45	61	15	13

Task 2 was divided into two parts. In the first part, participants were asked to add one a la carte item and two specific meal plans at their own pace, allowing them to freely explore the app in a "free play" mode. This approach provided a better understanding of how users naturally interact with the system without constraints. In the second part, participants were instructed to adjust their shopping cart by removing a specific meal plan.

On average, participants took 2 minutes and 30 seconds to complete the task, requiring an average of 24.3 clicks with an error rate of 10.7 clicks. Overall, 55% (5 out of 9) participants successfully completed the task. With nearly half of participants unable to complete the task successfully, all exhibited signs of frustration, cognitive overload, and stress. Frequent backtracking and the need to remember ingredient details created significant friction in the shopping experience. As one participant (P5) stated, "I was tempted to just clear out my cart and restart, that would've been easier."

Task 3: Finding Membercard

	P1	**P2**	**P3**	**P4**	**P5**	**P6**	**P7**	**P8**	**P9**
Task on Time	1:52	11s	15s	8s	47s	55s	12s	26s	15s
Completion Success?	Y	Y	Y	Y	Y	Y	Y	Y	Y
Error Rate (#) **Each tap the user did that was not directed towards goal**	8	0	1	0	2	9	0	0	0
Number of Clicks	11	2	3	2	4	11	2	2	2

Task 3 was the easiest task according to the data. During this task, participants were asked to locate their member card for in-person purchases. On average, participants completed the task in 33.4 seconds, requiring an average of 4.3 clicks with an error rate of 2.2 clicks. All participants successfully completed the task with almost no issues. However, some noted that while the process was smooth, it could be faster and placed in a more intuitive location. As one participant (P5) stated, “Having the member card in the wallet doesn’t make sense because it’s technically not linked to a bank or payment method.”

Suggested Usability Enhancement (Giovanni & Zaijing)

The goal of conducting the interface and usability evaluation was to identify opportunities for improvement to inform the redesign of the Safeway App. After reviewing the findings holistically across both interface and usability data, the team developed the following recommendations:

Clipping & Applying Coupons

1. **Offer Informative Feedback:** Provide clear, immediate feedback when users tap “Clip Deal” by displaying a confirmation state such as “Successfully Clipped.”
2. **Design Dialogue to Yield Closure:** Introduce a contextual next step to guide users through the couponing process. Once a coupon is clipped, the button should change from “Clip Deal” to “Shop Deal” to communicate task completion and prompt the next action.

Shopping Cart

1. **Affordance:** Replace the standalone numeral with clear quantity adjustment controls, such as +/– buttons, to align with established interaction patterns and affordances
2. **Efficiency of Use:** Integrate an X icon for item removal or a checkbox to support multi-item deletion. This reduces ambiguity, prevents errors, and creates a more intuitive and efficient cart management experience
3. **Keep Users in Control:** Make the address element interactive to allow users to change the location of their designated store within the shopping cart, and to prevent backtracking
4. **Visibility of System Status & Error Prevention:** Provide immediate feedback on product availability to avoid frustration and backtracking
5. **Gestalt Proximity:** Integrate a tab button to allow users to view meal plans and its nested items separately to provide efficient management

Member Card(QR code)

1. **Move member QR code from wallet to member page:** Relocate the QR code to a more accessible location, so users can easily find it without navigating multiple layers.

2. **Unify the design and context of QR code:** Standardize the design of the member card (QR code) from the U deal page and the member card (QR code) in the member section so they share the same visual style and contextual meaning.

Search Bar

1. **Keep Users in Control, Cater to Universal Usability, Affordance:** Improve the search algorithm to provide support partial matches and predictive suggestions
2. **Provide immediate feedback:** Provide real time feedback such as “no exact matches found” or “did you mean” to guide users and maintain trust in the system.

Conclusion (Giovanni)

In conclusion, the study identified that while the Safeway App provides a minimalist design, its inconsistent shopping flow and violation of design principles creates extensive friction in the user’s shopping experience. After a final assessment of the UI evaluation, survey, and usability data, the team determined that the Safeway App’s core challenges stem from unclear feedback, inefficient task flows, and inconsistent interface patterns. The recommendations provided aim to improve clarity, efficiency, and user control across key features.

For a future study, the team would conduct a cognitive walkthrough to further triangulate our understanding of user experiences. A cognitive walkthrough is a method for evaluating the user interface of products by analyzing the mental processes required by users. Its focus is on the learnability of an application, and is guided by (3) research questions.

Our study was guided by the following 3 research questions, **RQ1:** How effectively can customers locate, clip, and apply coupons, and find eligible items to add to their cart? **RQ2:** What challenges do customers face when managing their shopping carts when adding a la carte items and meal plans? **RQ3:** How easily can customers access their member card to scan during in-store purchases, and how does this process influence their overall satisfaction with the app? What we can infer is that, many users can generally locate, clip, and apply coupons, however with the tradeoff of consistent breakdowns of efficiency and unclear feedback. In regards to RQ2, all users faced cognitive overload and intense frustration due to inconsistent grouping. For RQ3, users were able to access their membercard seamlessly and were confident when doing the task. However, half of participants wished it was still a quicker process since the feature is important to many Safeway customers.

Key Takeaways (Everyone)

1. **Usability testing can make users reveal real behavior:** we observe participants have other issues which we haven't found in heuristic evaluation like the hesitation with whether they successfully remove all ingredients from one meal plan. And many people are frustrated with search functions.

2. **Survey and Questionnaire can make us learn more about users:** Although we already use usability tests to learn real behavior, the survey and questionnaire findings can complement our understanding. According to these, we understand the coupon function is the target user's motivation to use this app.

3. **User motivation is based on their savings:**
   Our research showed that the majority of users use the Safeway app to shop for deals as well as to browse products in general. This finding suggests that user engagement is dependent on coupon visibility and reliability, and that Safeway and its users would benefit greatly if this feature was more intuitive and easy to use.

4. **Small design inefficiencies accumulate into major user frustration.**
   Compounding issues such as unclear coupon feedback, extra navigation steps, and rigid cart editing led to measurable delays and frustration during testing. These small moments of friction consistently led to a reduction in satisfaction and prevented users from using the app smoothly without hesitation.

5. **Triangulation is Key:** Employing multiple research methods to converge on a focused set of insights is essential for developing a holistic understanding of user behavior and needs. Combining qualitative and quantitative approaches strengthens the validity of findings and enhances the overall credibility of the research.

6. **Insights = Goals + Needs in Design Language:** After conducting usability testing, it's essential to translate collected data into actionable insights. Researchers need to bridge user needs and pain points into clear design opportunities that allow designers to make meaningful, evidence-based improvements to the product.

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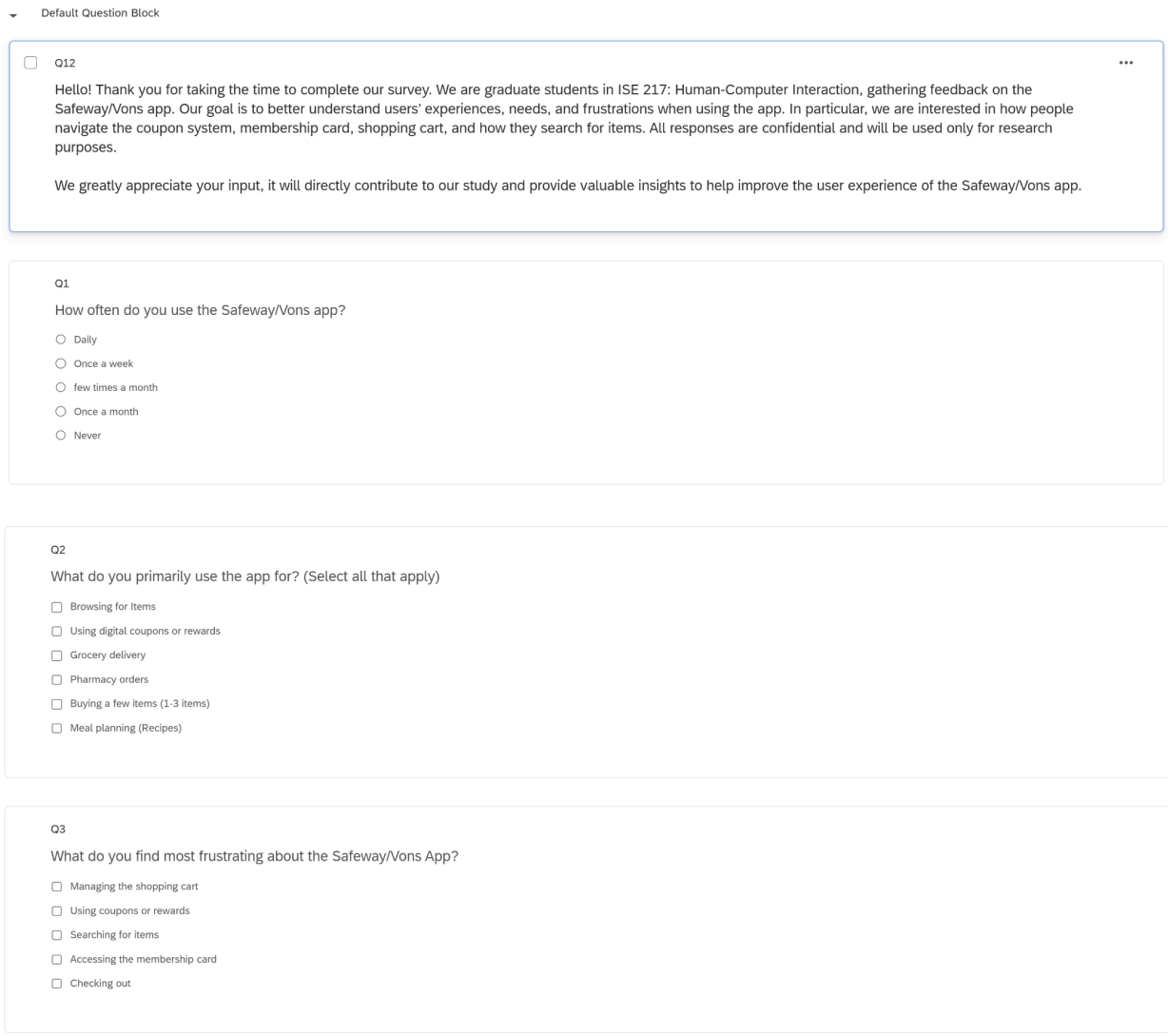
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Moallem, A. (2025). *Affordances* (pp.7-15). San Jose State University

Appendix (Team)

Survey



Q4

How important are deals and coupons to you when shopping for groceries?

- ○ Not at all important
- ○ Slightly important
- ○ Moderately important
- ○ Very important
- ○ Extremely important

Q5

How would you rate the ease or difficulty of finding and applying coupons?

- ○ Very easy
- ○ Easy
- ○ Neutral
- ○ Difficult
- ○ Very Difficult

Q6

When shopping for groceries, who are you usually shopping for?

- ○ Just myself
- ○ Myself and one other person
- ○ A household of 3 or more people

Q7

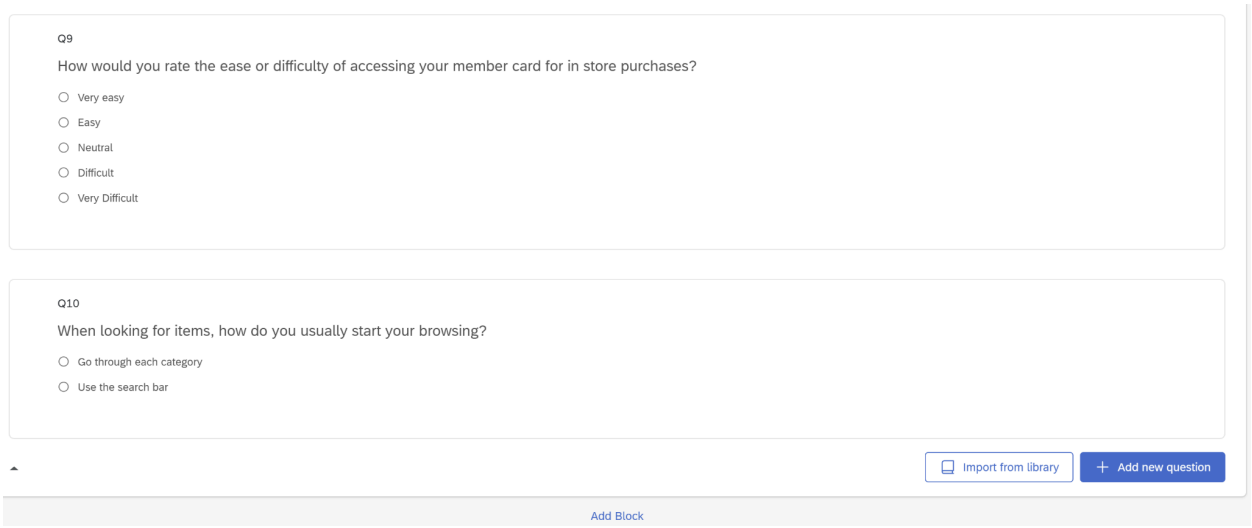
How often do you remove items from your shopping cart before checkout?

- ○ Never
- ○ Rarely
- ○ Sometimes
- ○ Often
- ○ Always

Q8

How would you rate the ease or difficulty of removing items from the shopping cart?

- ○ Very Easy
- ○ Easy
- ○ Neutral
- ○ Difficult
- ○ Very Difficult



Usability Testing Script

Introduction

Welcome to our study, thank you for volunteering to be a part of this study today. My name is _____ and these are my colleagues _____ who will be conducting the study with me.

Consent

Before we begin, we will require your consent for this study. Please read over and sign this document, let me know if you have any questions or concerns.

Study Overview

Welcome to our study, today we are going to have you perform tasks with the Safeway App. The Safeway App is a grocery store app, where shoppers can buy many necessities such as food and medicine. The purpose of this study is to evaluate the general usability of certain features of the app.

This study will take about 30 - 45 minutes to complete and will include us providing you with some tasks and goals to complete using the Safeway App. During a task, please refrain from asking any questions regarding the tasks since the goal is to create a realistic scenario of how you would use it. As you perform the tasks, **please remember that this is not a test of you, but rather of the app so don't be afraid to do what you think is right.**

Any questions?

Pre-Test Questions

First we will begin by asking some questions regarding your grocery app usage.

1. How often do you shop online for food or daily necessities? (Answer like: more than once a week/ Once a week/ Once a month)
    a. Once a week
2. Have you used the Safeway app or other similar shopping apps? If yes, which one?
    a. Yes safeway and whole foods
3. Do you usually look for digital coupons or promotions when you shop online?
    a. yes
4. What is the most common frustration you face when you shop for food online?
    a. Safeway – doenst load quickly
    b. Search doesnt work and location address doesnt work
    c. Lack of feedback on stock
    d. coupon

Thank you for answering those questions , now we will start the study on the Safeway App.

Feature #1: Clipping and Applying Coupons to Eligible Items

Task
Moderator: You downloaded the Safeway App to find good deals on groceries. Find any coupon you want to use and successfully apply it. **Task Navigation Flow:** 1. Open Safeway App 2. Navigate to for for U Deals screen (click either on icon or navigation button) 3. Selected desired coupon 4. Navigate to “Clipped” section in the For U Deals section 5. Select desired coupons 6. Navigate to “clipped” section 7. Click on the coupon 8. Click on “eligible items” 9. Choose one of the eligible items 10. Add to cart
Perception Probe (ask when participant barely clips the coupon) #1 :

Do you think the discount will be applied once you clip it?

Performance Observations	
Time on Task	
Completion Success	
Error Rate (#) **(When they clip it , but don't go straight to "clipped" , every action is considered an error)**	
Number of Clicks (Navigation Efficiency)	

Behavioral Observations (Frustration, Hesitation, Confusion, Facial expressions, Body Posture)

Subjective
Do you think the discount will be applied once you clip it?

Post - Task Questions	Responses
Q1: (Ease of Use) On a Scale of 1 – 5 , how easy or difficult was it to find and clip a coupon to an eligible	

item? 1 = Very Easy 2 = Easy 3 = Neutral 4 = Difficult 5 = Very Difficult	
Q2: Did you feel confident that the coupon was successfully clipped and applied? Why or why not?	
Q3: Did anything about the process feel frustrating or confusing? If yes, what specifically?	
Q4: What would have made the process easier for you?	

Feature #2: Reviewing & Adjusting Shopping Cart

Task 2.1 (Free Play)
Moderator: Your friend told you about some good items and you went shopping for groceries in the Safeway App. You need some a la carte items and want to try one of their meal plans from their meal hub that your friend mentioned. **Add 1 a la carte item and 2 meal plans**. **Add a banana and add the Almond - Crusted Turkey Patties with Garlic Green Beans and Cilantro lime chicken thighs.** Please let me know when you have added them to your cart. **Task Navigation Flow:** 1. Open Safeway App 2. Scroll down on home screen 3. Add Banana 4. Click “My Lists” 5. Click Meals Hub 6. Scroll down until you see “Budget friendly dinners” 7. Add Almond chicken 8. Scroll down to “budget friendly” 9. Add Cilantro Lime Chicken Thighs

10. Add to Cart

Note for moderator: If they can't find a meal hub for meal plans , assist them to add the 2 meal plans since we are not testing this aspect until they reach task 2.2.

Task 2.2
Moderator: When reviewing your shopping cart you realized you're over budget. Please remove all ingredients from the Cilantro - Lime Chicken Thighs Meal Plan **Task Navigation Flow:** 1. Remove Avacado 2. Remove Chicken thighs 3. Remove Cilantro 4. Remove Garlic 5. Remove Grape Tomatoes 6. Remove Lime 7. Remove Red Onion 8. Remove Black Pepper 9. Remove Extra Virgin Olive Oil 10. Remove Salt
Perception Probe (ask when participant barely clips the coupon) #1 : Do you think you removed all the ingredients for the meal plan?

Performance Observations	
Time on Task	
Completion Success	
Error Rate (#) **(Each time they go back to the meal plan page to check the ingredients)**	
Number of Clicks (Navigation Efficiency)	

Behavioral Observations (Frustration, Hesitation, Confusion, Facial expressions, Body Posture)

Subjective
Do you think there is a way to see the list of ingredients for each meal plan?

Post - Task Questions	Responses
Q1: On a Scale of 1 – 5 , how easy or difficult was to remove all the items for a specific meal plan? 1 = Very Easy 2 = Easy 3 = Neutral 4 = Difficult 5 = Very Difficult	
Q2: On a scale of 1 - 5, how confident did you feel in completing the task? 1 = Not at all confident 2 = Slightly confident 3 = Moderately confident 4 = Very Confident 5 = Extremely Confident	
Q3: Did anything about the process or experience feel frustrating or confusing? If yes, what specifically?	
Q4: What would have made the process easier for you?	

Q5: On a scale of 1 - 5, rate your level of satisfaction with deleting items from meal plans in the shopping cart? 1 – Very dissatisfied 2 – dissatisfied 3 – unsure 4 – satisfied 5 – Very satisfied	

Feature #3: Finding your Member Card

Task
Moderator: You went in-person to do shopping at Safeway. You remembered you clipped a coupon and want to make sure you get your points and discounts. Find your member card. **Task Navigation Flow:** Method 1 Tasks: 1. Open Safeway App 2. Click on for UDeals 3. Click on “Clipped” tab 4. Click on bar code UI element at bottom of screen Method 2 Tasks: 1. Open Safeway App 2. Click on member navigation button 3. Click on Wallet

Performance Observations	
Time on Task	
Completion Success	
Error Rate (#) **(When they go anywhere besides the U deals page or**	

member page)	
Number of Clicks (Navigation Efficiency)	

Behavioral Observations (Frustration, Hesitation, Confusion, Facial expressions, Body Posture)

Subjective
Do you think it was a quick process?

Post - Task Questions	Responses
Q1: On a Scale of 1 – 5 , how easy or difficult was it to find your member card 1 = Very Easy 2 = Easy 3 = Neutral 4 = Difficult 5 = Very Difficult	
Q2: On a scale of 1 - 5, how confident did you feel in completing the tasks? 1 = Not at all confident 2 = Slightly confident 3 = Moderately confident 4 = Very Confident 5 = Extremely Confident	

Q3: Did anything about the process or experience feel frustrating or confusing? If yes, what specifically?	
Q4: What would have made the process easier for you?	
Q5: On a scale of 1 - 5, rate your level of satisfaction with finding your membercard 1 – Very dissatisfied 2 – dissatisfied 3 – unsure 4 – satisfied 5 – Very satisfied	

Thank you for participating in our study, now we will ask you some post study questions about your overall experience

Post – Study Questions	
Q1: On a scale of 1 - 5, how satisfied are you with your overall experience using the Safeway App? 1 – Very dissatisfied 2 – dissatisfied 3 – unsure 4 – satisfied 5 – Very satisfied	
Q2: How user friendly did you find the Safeway App? Very User Friendly Somewhat user friendly Neutral Somewhat Difficult	

Very Difficult	
Q3: Would you recommend the Safeway App to a friend? Yes No Maybe	
Q4: What did you find most frustrating or confusing with your experience with the Safeway App?	
Q5: Do you have any suggestions for improving the Safeway App?	